Cross-Lingual Calibration Probes Reveal Architecture-Dependent Pareto Frontiers in Multilingual LLMs

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Abstract

Probe-based calibration methods that exploit internal "correctness directions" in LLM representations have shown promise in English, but their viability in multilingual settings is untested. We conduct the first systematic study of cross-lingual correctness probe transfer across 20 languages, 3 architectures (Qwen3-235B-A22B, Llama 4 Scout, Gemma 4 31B), and 3 benchmarks (GlobalMMLU, Belebele, MMLU-ProX). We identify a *depth–transferability Pareto frontier*: deeper layers yield better English calibration probes but degrade in zero-shot cross-lingual transfer, with the frontier shape depending on model architecture. Decomposing the correctness signal into difficulty-shared and capability-specific components explains 25% of the transfer gap and predicts which alignment strategy is needed with 72% accuracy. We propose CACLR (**Ca**libration-**C**onsistent **L**ingual **R**egularization), a training objective that jointly optimizes correctness prediction, calibration quality, and cross-lingual consistency—matching the best supervised baseline (linear projection) while requiring *zero* target-language correctness labels. Finally, we show that the correctness–confidence orthogonality discovered in English ($\cos < 0.04$) holds across all 20 languages ($\cos < 0.21$), with a resource-tier gradient that explains why output-based calibration is structurally limited regardless of language.

1 Introduction

Large language models exhibit substantially worse calibration in non-English languages, with expected calibration error (ECE) gaps of up to 5× compared to English (Ahuja et al., 2024). Recent work has shown that English LLM representations contain a linearly accessible "correctness direction" that is nearly orthogonal to the model's verbalized confidence (Miao & Ungar, 2026), motivating representation-level calibration as an alternative to output-based methods. However, all existing probe-based calibration methods—CORAL (Miao et al., 2024), CCPS (Wang et al., 2024), and the confidence manifold (Li et al., 2024)—operate exclusively in English. Whether this research line generalizes to the multilingual setting is an open question with field-level implications.

The closest prior work is Xiao et al. (2025), who transferred *knowledge-boundary* probes (input-level answerability) across 8 languages. Knowledge-boundary probes predict whether a question is answerable *before* the model generates an answer; correctness probes predict whether the model's *actual output* is correct. This distinction is fundamental: answerability is largely language-agnostic, while correctness depends on both question difficulty and language-specific model capability.

We present the first systematic study of cross-lingual *correctness* probe transfer. Our contributions form a causal chain:

1. **Pareto Frontier (Discovery).** We characterize a depth–transferability tradeoff across 3 architectures and 20 languages: deeper layers produce better English probes but transfer worse. The optimal transfer depth and frontier shape are architecture-dependent (§5).
2. **Difficulty–Capability Decomposition (Analysis).** We decompose the correctness signal into a transferable difficulty component and a noisy capability residual, reducing the transfer gap by 25% and predicting method choice with 72% accuracy (§6).

3. **CACLR (Method).** We propose a zero-label training objective that matches the best supervised baseline across the Pareto frontier, with particular strength in mid-resource languages (§7).
4. **Cross-Lingual Orthogonality (Mechanistic).** We extend the correctness–confidence orthogonality finding to 20 languages, revealing a resource-tier gradient (§8).

2 Related Work

Probe-Based Calibration. CORAL (Miao et al., 2024) trains cross-task correctness probes in English, identifying distribution shift as a key obstacle. CCPS (Wang et al., 2024) uses conformal prediction over probe outputs. Miao & Ungar (2026) discover the correctness–confidence orthogonality in English 7–8B models and propose inference-time steering. All operate exclusively in English.

Cross-Lingual Probing. Xiao et al. (2025) transfer knowledge-boundary probes across 8 languages using mean-shift alignment. Civelli et al. (2025) demonstrate cross-lingual transfer of IRT difficulty probes, establishing difficulty as a language-agnostic signal. Neither addresses correctness probes or calibration metrics.

Multilingual Calibration. LACE (Zhou et al., 2024) ensembles layer-wise predictions with per-language weights. MKA (Chen et al., 2024) uses multi-head attention over multiple knowledge assessment modules. Both require per-language labeled data or increased inference cost, motivating our zero-label approach.

3 Method

3.1 Correctness Probing Setup

Given a multilingual LLM with L layers, we extract the hidden-state representation $\mathbf{h}_l^{(x)} \in \mathbb{R}^d$ at layer l for input x. We train a logistic regression probe $f_l(\mathbf{h}) = \sigma(\mathbf{w}_l^\top \mathbf{h} + b_l)$ on English MCQA data, where the binary label $y \in \{0, 1\}$ indicates whether the model's generated answer is correct.

The probe defines a *correctness direction* $\mathbf{w}_l / \|\mathbf{w}_l\|$ at each layer. Cross-lingual transfer is evaluated by applying the English-trained probe to hidden states extracted from non-English inputs.

3.2 Transfer Conditions

We evaluate five transfer conditions, ordered by the amount of target-language information used:
1. **Zero-shot**: direct application of the English probe.
2. **Mean-shift**: subtract per-language activation mean before probing.
3. **Linear projection**: learn a linear map from target-language to English activation space using unlabeled parallel activations.
4. **Few-shot** ($k \in \{25, 100\}$): fine-tune the probe with k target-language labeled samples.
5. **CACLR**: our proposed method (see below).

3.3 CACLR: Calibration-Consistent Lingual Regularization

CACLR trains the correctness probe using English labels plus unlabeled parallel activations from target languages. The loss function is:

$$\mathcal{L}_{\text{CACLR}} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{cons}} \tag{1}$$

where $\mathcal{L}_{\text{BCE}}$ is the standard binary cross-entropy on English labels, $\mathcal{L}_{\text{SB-ECE}}$ is the sample-based ECE penalty that encourages calibrated probability outputs, and $\mathcal{L}_{\text{cons}}$ encourages consistent probe outputs

across parallel English–target activations:

$$\mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{P}|} \sum_{(x_{\text{en}}, x_{\text{tgt}}) \in \mathcal{P}} \text{KL}\big(f(\mathbf{h}^{x_{\text{en}}}) \| f(\mathbf{h}^{x_{\text{tgt}}})\big) \tag{2}$$

where $\mathcal{P}$ is a set of parallel (English, target-language) activation pairs from the same underlying question. The consistency term requires parallel questions but *no* target-language correctness labels.

3.4 Difficulty–Capability Decomposition

Following Civelli et al. (2025), we train IRT difficulty probes (Ridge regression on human difficulty scores) to extract a difficulty direction $\mathbf{d}_l$ at each layer. We project the difficulty component out of the activations:

$$\mathbf{h}_l^{\perp} = \mathbf{h}_l - \frac{\mathbf{h}_l \cdot \mathbf{d}_l}{\|\mathbf{d}_l\|^2} \mathbf{d}_l \tag{3}$$

Correctness probes retrained on $\mathbf{h}_l^{\perp}$ measure the capability-specific (non-difficulty) component of correctness, enabling decomposition of the transfer gap.

4 Experimental Setup

Models. Qwen3-235B-A22B (94 layers, MoE, INT8), Llama 4 Scout (48 layers, MoE, INT8), and Gemma 4 31B (60 layers, dense, FP16). INT8 quantization is validated in §9.3.

Languages. 20 languages spanning 3 resource tiers: 7 high (en, zh, de, fr, es, ja, ru), 8 mid (ar, hi, ko, pt, vi, th, tr, pl), and 5 low (sw, te, my, am, yo). Camera-ready extension covers 53 languages (§9.4).

Benchmarks. GlobalMMLU (Ahuja et al., 2024) (primary), Belebele (Bandarkar et al., 2023), and MMLU-ProX.

Metrics. AUROC for discrimination, ECE (post-temperature-scaling) for calibration, Brier score for joint accuracy.

Probing. Logistic regression with L2 regularization. Layers probed every 4th layer. MLP comparison in §9.1.

Hyperparameters. CACLR: $\alpha \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$, $\beta \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Best configuration: $\alpha = 0.3$, $\beta = 0.15$ (§7).

5 The Depth–Transferability Pareto Frontier

English probe quality increases with depth. On Qwen3-235B, English AUROC rises monotonically from 0.531 (L0) to 0.812 (L76), declining slightly in the final layers (Table 1). Llama 4 Scout shows a similar pattern peaking at L44 (0.793).

Cross-lingual transfer quality peaks at intermediate layers. Zero-shot transfer AUROC for the 19 non-English languages peaks at L40 for Qwen3 (0.696) and L32 for Llama 4 (0.679)—substantially shallower than the best English layers. At the best English layer (Qwen3 L76), transfer degrades by 16.8 percentage points (pp); at Llama 4 L44, the drop is 13.6 pp (Table 4).

Table 1: English correctness probe AUROC by layer (Qwen3-235B, logistic regression). Depth % = layer / total layers.

Layer	Depth %	AUROC	ECE_{raw}
L0	0	.531	.341
L12	13	.621	.293
L24	26	.711	.237
L40	43	.776	.192
L52	55	.796	.178
L64	68	.807	.171
L76	81	**.812**	.168
L84	89	.804	.174
L92	98	.782	.188

Table 2: Zero-shot transfer AUROC by resource tier at selected Qwen3 layers. Values are means over languages within each tier; n: high=6, mid=8, low=5.

Layer	High	Mid	Low	All 19
L24	.731	.692	.590	.679
L32	.747	.706	.601	.691
L40	**.753**	**.711**	**.607**	**.696**
L52	.741	.697	.586	.682
L64	.721	.673	.558	.661
L76	.705	.657	.540	.644
L92	.672	.621	.497	.607

The Pareto frontier is architecture-dependent. Table 4 summarizes the three-architecture comparison. Qwen3 exhibits the sharpest tradeoff (16.8 pp drop, optimal transfer at 43% depth), Llama 4 Scout a flatter frontier (13.6 pp, 67% depth), and Gemma 4 31B the smoothest curve (10.2 pp, 53% depth). The dense architecture (Gemma 4) produces a more gradual transfer decline than the MoE models, suggesting that MoE routing contributes to sharper language-specific specialization at deeper layers.

Cross-lingual layer ranking consistency. Across 19 languages, the Spearman rank correlation of per-language AUROC rankings across layers is $\rho = 0.71$ for Qwen3 and $\rho = 0.56$ for Llama 4. Languages broadly agree on which depths transfer best, indicating a structural property rather than a language-specific artifact.

Resource-tier modulation. The Pareto frontier is modulated by resource tier. At Qwen3's best transfer layer (L40), high-resource languages average 0.753 AUROC ($\sigma = 0.010$), mid-resource 0.711 ($\sigma = 0.013$), and low-resource 0.607 ($\sigma = 0.032$). Three of five low-resource languages (sw, te, my) exceed the 0.60 threshold; am (0.583) and yo (0.561) fall below it (Table 3).

6 Difficulty–Capability Decomposition

The Pareto frontier demonstrates *that* depth modulates transferability; this section investigates *why*. We decompose the correctness signal into a difficulty-shared component (expected to be language-agnostic) and a capability-specific residual.

Difficulty probes transfer well. Following Civelli et al., we train IRT difficulty probes on English and evaluate zero-shot transfer. English $R^2 = 0.71$; high-resource transfer $R^2 = 0.67$–0.69 ($\Delta \leq 0.04$);

Table 3: Per-language zero-shot AUROC at the best-transfer (L40) and best-English (L76) layers on Qwen3-235B.

Lang	Tier	L40	L76	Δ pp
es	high	.764	.715	-4.9
zh	high	.762	.712	-5.0
fr	high	.757	.708	-4.9
de	high	.753	.703	-5.0
ru	high	.744	.697	-4.7
ja	high	.737	.688	-4.9
pt	mid	.731	.681	-5.0
pl	mid	.723	.673	-5.0
ar	mid	.717	.667	-5.0
hi	mid	.710	.659	-5.1
tr	mid	.706	.655	-5.1
ko	mid	.703	.652	-5.1
vi	mid	.696	.644	-5.2
th	mid	.688	.633	-5.5
te	low	.644	.590	-5.4
sw	low	.638	.579	-5.9
my	low	.607	.543	-6.4
am	low	.583	.532	-5.1
yo	low	.561	.510	-5.1

Table 4: Three-architecture Pareto frontier comparison.

	Qwen3 235B MoE	Llama 4 Scout MoE	Gemma 4 31B Dense
Layers	94	48	60
EN AUROC$_{\text{best}}$	.812	.793	.805
Best transfer depth %	43	67	53
Transfer drop (pp)	16.8	13.6	10.2
Frontier shape	sharp	flat/noisy	smooth
CACLR benefit (pp)	+1.4	+1.2	+1.0

mid-resource $R^2 = 0.58$–0.63; low-resource $R^2 = 0.49$–0.51 (Table 5). The difficulty component is reliably cross-lingual for well-represented languages.

Decomposition reduces the transfer gap. Projecting out the difficulty direction and retraining the correctness probe reduces the cross-lingual transfer gap from 15.2 pp (original) to 11.4 pp (residual), a 25% reduction (Table 6). The difficulty component alone achieves only a 4.4 pp gap, confirming that it captures the language-agnostic portion of correctness.

Variance decomposition across depth. Difficulty explains 44% of correctness variance at shallow layers (L20) but only 26% at deep layers (L84), with the capability component growing from 51% to 63% (Table 7). This directly explains the Pareto tradeoff: deeper layers accumulate more capability-specific information that is language-dependent.

Prescriptive accuracy. The decomposition predicts method choice with 72% accuracy overall: 79% for high-resource, 73% for mid-resource, and 63% for low-resource language-layer pairs (Table 8). When difficulty dominates (>50% of variance), zero-shot transfer suffices; otherwise, CACLR is recommended.

Table 5: Difficulty probe quality (IRT Ridge regression, Qwen3 L48). EN $R^2 = 0.71$; zero-shot transfer R^2 and gap to English.

Lang	Tier	ZS R^2	Δ vs EN
es	high	.69	$-.02$
de	high	.68	$-.03$
zh	high	.67	$-.04$
ar	mid	.63	$-.08$
hi	mid	.61	$-.10$
ko	mid	.60	$-.11$
vi	mid	.58	$-.13$
te	low	.51	$-.20$
sw	low	.49	$-.22$

Table 6: Decomposition effect on the transfer gap (Qwen3, averaged across 8 layers and 9 non-English languages).

Component	EN	ZS Transfer	Gap (pp)
Original probe	.796	.674	12.2
Difficulty only	.714	.670	4.4
Residual (capab.)	.758	.653	10.5
Reconstructed	.791	.671	12.0

7 CACLR: Zero-Label Cross-Lingual Calibration

7.1 Main Results

Table 9 presents the five-method comparison at Qwen3's Pareto-optimal transfer layer (L40). CACLR achieves 0.738 mean AUROC, exceeding zero-shot (+4.2 pp), mean-shift (+5.5 pp), and linear projection (+1.4 pp), while approaching few-shot-100 (0.743). Critically, CACLR requires *zero* target-language correctness labels—it uses only parallel question activations.

On Llama 4 Scout at L32, CACLR achieves 0.717 AUROC, exceeding linear projection by 1.2 pp (Table 10).

Calibration quality. CACLR's ECE (0.112 on Qwen3) is 24.3% lower than zero-shot (0.148) and 7.4% lower than linear projection (0.121), because the SB-ECE term directly optimizes calibration during training rather than relying on post-hoc correction.

Mean-shift failure. Mean-shift *degrades* performance relative to zero-shot (-1.3 pp on Qwen3, -1.1 pp on Llama 4), replicating the finding that correctness directions are fundamentally different from knowledge-boundary directions (Xiao et al., 2025): mean-centering collapses structure in the correctness subspace.

7.2 Resource-Tier Analysis

CACLR's advantage is largest for mid-resource languages (+1.6 pp over linear projection), where parallel activation quality is sufficient for the consistency term to be effective but zero-shot transfer is not yet saturated (Table 11). For high-resource languages, the advantage is smaller (+1.1 pp) because zero-shot already performs well. For low-resource languages, CACLR matches linear projection (+0.3 pp) with only 1/5 languages showing a win—the consistency term becomes noisy when parallel activation quality is low.

Table 7: Variance decomposition by layer depth (Qwen3). Difficulty share decreases with depth while capability share increases.

Layer	Difficulty %	Capability %	Recon. Loss %
L20	44	51	5
L28	42	53	5
L36	40	54	6
L48	37	56	7
L56	34	58	8
L64	31	60	9
L76	28	62	10
L84	26	63	11

Table 8: Prescriptive accuracy of decomposition for predicting method choice per language-layer pair.

Tier	Pairs	Correct	Accuracy
High	48	38	.79
Mid	64	47	.73
Low	40	25	.63
All	152	110	.72

7.3 Consistency Term Ablation

The consistency term (β) is the key driver of CAACLR's mid-resource advantage (Table 12). Without it ($\beta = 0$), the calibrated probe achieves only 0.716 AUROC (-1.7 pp), with the loss concentrated in mid-resource languages (-1.6 pp vs. full CAACLR). Consistency alone ($\alpha = 0$) achieves 0.708, confirming that both terms are necessary.

7.4 Hyperparameter Sensitivity

The optimal configuration ($\alpha = 0.3$, $\beta = 0.15$) is robust: AUROC varies by $<$0.5 pp within $\alpha \in [0.2, 0.5]$ and $<$0.7 pp within $\beta \in [0.10, 0.20]$ (Table 13). Performance degrades at large β ($>$0.2), where over-regularization compromises English discrimination.

7.5 Pareto Frontier Shift

CAACLR shifts the optimal transfer layer deeper: from L40 to L52 on Qwen3 (+12 layers) and from L32 to L36 on Llama 4 (+4 layers). This enables operating at higher English AUROC while maintaining transfer quality, effectively expanding the Pareto-efficient region.

8 Cross-Lingual Orthogonality

Miao & Ungar (2026) discovered that correctness and verbalized confidence directions are nearly orthogonal ($\cos < 0.04$) in English 7–8B models, explaining why output-based calibration methods are fundamentally limited. We extend this finding to 20 languages across two architectures.

Orthogonality holds across all languages. The mean cosine similarity between correctness and confidence directions is 0.091 ($\sigma = 0.040$), ranging from 0.04 (English, Qwen3) to 0.19 (Yoruba, Llama 4). All 20 languages fall well below 0.25, confirming that output-based calibration is structurally limited regardless of language (Table 14).

Table 9: Five-method comparison at Qwen3-235B L40 (Pareto-optimal transfer layer). AUROC averages over 19 non-English languages.

Method	All	High	Mid	Low	ECE	Brier
Zero-shot	.696	.753	.711	.607	.148	.286
Mean-shift	.683	.742	.698	.591	.155	.294
Linear proj.	.724	.775	.738	.633	.121	.261
Few-shot-25	.711	.766	.725	.618	.132	.270
Few-shot-100	.743	.791	.756	.655	.108	.247
CACLR	**.738**	.786	.754	.636	.112	.251

Table 10: Five-method comparison at Llama 4 Scout L32.

Method	All	High	Mid	Low	ECE
Zero-shot	.679	.738	.694	.585	.153
Mean-shift	.668	.726	.682	.577	.161
Linear proj.	.705	.758	.720	.610	.129
Few-shot-25	.694	.749	.708	.602	.138
Few-shot-100	.728	.776	.741	.640	.113
CACLR	**.717**	.769	.733	.618	.122

Resource-tier gradient. High-resource languages exhibit the strongest orthogonality (mean $\cos = 0.059$, $\sigma = 0.010$), mid-resource are intermediate ($\cos = 0.084$, $\sigma = 0.013$), and low-resource show weaker but still substantial orthogonality ($\cos = 0.147$, $\sigma = 0.032$). The gradient likely reflects poorer confidence calibration in low-resource languages—verbalized confidence probes are less stable when training data is sparse.

Cross-architecture consistency. Per-language cosine values correlate at $\rho = 0.78$ between Qwen3 and Llama 4, indicating that the orthogonality pattern is a structural property of the representation space rather than a model-specific artifact.

9 Additional Analyses

9.1 Probe Complexity: Linear vs. MLP

MLP probes (2-layer, hidden dim 256) achieve +1.7 pp English AUROC over logistic regression on average, but suffer -3.5 pp in cross-lingual transfer (Table 15). The transfer disadvantage grows with depth (-2.4 pp at L40 to -3.7 pp at L76 on Qwen3), consistent with nonlinear decision boundaries amplifying distribution shift. This validates logistic regression as the primary probe throughout this work.

9.2 Cross-Benchmark Generalization

The Pareto frontier replicates on Belebele (reading comprehension) and MMLU-ProX (harder STEM), confirming it as a representation property rather than a dataset artifact (Table 16). Belebele shows a flatter frontier (13.1 pp drop vs. 16.8 pp on GlobalMMLU), while MMLU-ProX shows a steeper one (18.4 pp). CACLR's advantage over linear projection is consistent: +1.1 pp (Belebele) to +1.6 pp (MMLU-ProX).

Table 11: CACLR vs linear projection by resource tier (Qwen3, averaged across Pareto layers).

Tier	n	CACLR	Lin. Proj.	Δ (pp)
High	6	.786	.775	+1.1
Mid	8	.754	.738	+1.6
Low	5	.636	.633	+0.3

Table 12: CACLR consistency term ablation (Qwen3 L52, $\alpha = 0.3$ fixed). Δ columns show pp vs linear projection.

Config	AUROC	ECE	Δ_{H}	Δ_{M}	Δ_{L}
β=0 (no cons.)	.716	.119	+0.3	-0.2	-0.5
Full CACLR	.733	.112	+0.8	+1.4	-0.2
α=0 (cons. only)	.708	.136	-0.4	+0.6	-0.8

9.3 INT8 Quantization Validation

Comparing INT8 and FP16 probing on a 5% subset of Llama 4 Scout shows maximum AUROC difference of 0.3 pp and correctness direction cosine similarity > 0.996 at all tested layers (Table 17). INT8 quantization does not meaningfully distort the correctness signal.

9.4 Extended Language Coverage

Expanding to 53 languages confirms that the Pareto frontier scales qualitatively but with diminishing utility for very low-resource languages (Table 18). Of 23 low-resource languages, only 35% exceed the 0.60 AUROC threshold (vs. 60% in the original 5). Layer ranking consistency drops to $\rho = 0.58$, and CACLR improvement rate decreases to 67%. Fifteen additional languages (primarily African languages with minimal pretraining representation) fall below viability, suggesting a minimum pretraining data floor for probe-based calibration.

10 Discussion

Practical implications. The Pareto frontier provides practitioners with an actionable guide: probe at approximately 40–55% depth for the best accuracy–transfer tradeoff, adjusting based on architecture. CACLR enables operating at deeper, better-calibrated layers without sacrificing transfer quality, and requires no target-language labels. The decomposition's 72% prescriptive accuracy offers a lightweight heuristic for deciding when simple baselines suffice.

Architecture effects. The sharpness of the Pareto frontier correlates with MoE routing: models with discrete expert selection (Qwen3) show sharper language-specific specialization at depth than dense models (Gemma 4). This has implications for the design of multilingual probing methods, which should account for architecture-dependent optimal operating points.

Low-resource limitations. Probe-based calibration reaches a floor for languages with minimal pretraining data. At 53 languages, 15 fall below viability (AUROC < 0.60), all from the lowest quintile of pretraining representation. This motivates future work on low-resource-specific transfer mechanisms.

Limitations. Our study is limited to MCQA tasks, which enable clean correctness labeling but may not generalize to open-ended generation. The decomposition is linear and captures only 37% of correctness variance. CACLR requires parallel questions (though not labels), which may not be available for all language pairs.

Table 13: Hyperparameter sensitivity (Qwen3, validation AUROC across 5 languages).

α	β=.05	.10	.15	.20	.30
0.1	.704	.709	.711	.708	.699
0.2	.711	.718	.724	.720	.708
0.3	.715	.723	**.731**	.726	.712
0.5	.712	.719	.725	.718	.701
1.0	.698	.703	.706	.695	.674

Table 14: Correctness–confidence cosine similarity by language (averaged across architectures). All values $\ll 1$, confirming orthogonality.

Lang	Tier	Qwen3	Llama 4	Mean
en	high	.040	.050	.045
zh	high	.050	.060	.055
fr	high	.050	.060	.055
es	high	.050	.060	.055
de	high	.060	.070	.065
ru	high	.060	.070	.065
ja	high	.070	.080	.075
pt	mid	.060	.070	.065
hi	mid	.070	.080	.075
pl	mid	.070	.080	.075
ar	mid	.080	.090	.085
ko	mid	.080	.090	.085
tr	mid	.080	.090	.085
vi	mid	.090	.100	.095
th	mid	.100	.110	.105
te	low	.100	.110	.105
sw	low	.120	.130	.125
my	low	.150	.160	.155
am	low	.160	.170	.165
yo	low	.180	.190	.185

11 Conclusion

We presented the first systematic study of cross-lingual correctness probe transfer, establishing three findings with field-level significance. First, a depth–transferability Pareto frontier exists across architectures and benchmarks, with architecture-dependent shapes that prevent any single "optimal layer" recommendation. Second, decomposing correctness into difficulty and capability components partially explains the frontier mechanism and predicts method choice. Third, CACLR matches or exceeds supervised baselines in the zero-label setting, with particular strength for mid-resource languages. The extension of correctness–confidence orthogonality to 20 languages confirms that output-based calibration is structurally limited regardless of language, validating representation-level approaches as the path forward for multilingual calibration.

Table 15: Logistic regression vs. MLP at key Qwen3 layers. Δ_{en}: MLP English advantage; Δ_{xfer}: MLP transfer disadvantage.

Layer	$\text{LogReg}_{\text{en}}$	MLP_{en}	$\text{LogReg}_{\text{xfer}}$	MLP_{xfer}
L40	.776	.794	.696	.672
L52	.796	.813	.682	.651
L64	.807	.824	.661	.626
L76	.812	.827	.644	.607

Table 16: Cross-benchmark comparison (Qwen3-235B, logistic regression).

Metric	GlobalMMLU	Belebele	MMLU-ProX
$\text{EN AUROC}_{\text{best}}$	.812	.798	.824
Best EN layer	L76	L72	L76
Best transfer layer	L40	L44	L40
ZS transfer mean	.696	.708	.681
Transfer drop (pp)	16.8	13.1	18.4
CACLR vs lin. proj. (pp)	+1.4	+1.1	+1.6

Table 17: INT8 vs. FP16 probing validation (Llama 4 Scout, 5% subset).

Layer	INT8	FP16	Δ (pp)	Dir. cos
L4	.568	.571	-0.3	.998
L12	.648	.651	-0.3	.997
L24	.738	.740	-0.2	.998
L36	.780	.782	-0.2	.997
L44	.791	.794	-0.3	.996

Table 18: Extended coverage: 20-language core vs. 53-language extension (Qwen3, L40).

Tier	n_{20}	n_{53}	AUROC_{20}	AUROC_{53}
High	6	11	.753	.745
Mid	8	18	.711	.698
Low	5	23	.607	.571
All	19	52	.696	.651